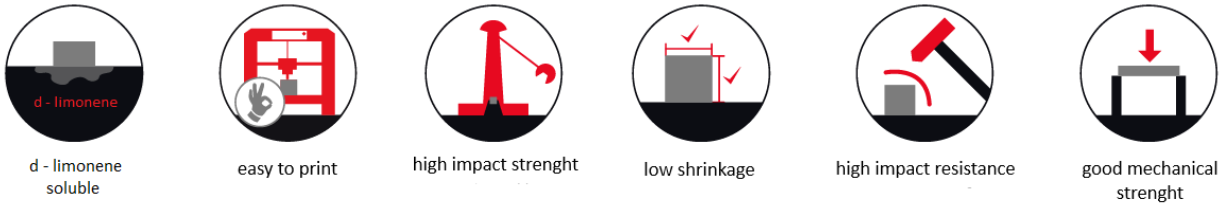
NOCTUO

PRODUCT CARD

HIPS Classic



1. GENERAL INFORMATION ABOUT THE PRODUCT

A high-impact polystyrene with low processing shrinkage that makes it possible to print large objects with thin walls. It is also an excellent support material for ABS - it prints under similar temperature conditions and dissolves d-limonene. HIPS Classic is not brittle at low temperatures and does not require drying before printing (it does not absorb water). Prints have a satin surface finish, are impact resistant and durable. They machine well.

Main HIPS Classic features:

- high impact strength,
- good mechanical strength,
- low processing shrinkage,
- reduced deformation when printing thin walls,
- perfect support for ABS,
- does not absorb water,
- soluble in d-limonene.

2. TECHNICAL PARAMETERS

CHARACTERISTICS	TEST METHOD	TEST CONDITIONS	IU	VALUE
	ISO			
PHYSICAL				
Density	ISO 1183	-	g/cm^3	1.04
Water absorption	ISO 62	23°C/24h	%	<0.1
MECHANICAL				
Yield strength	ISO 527	50mm/min	MPa	18
Elongation at break	ISO 527	50mm/min	%	55
Elastic modulus at tension	ISO 527	1mm/min	MPa	1700
Flexural strength	ISO 178	2mm/min	MPa	32

Notched impact strength, IZOD	ISO 180/4A	23°C-3,2mm	J/m	110
Notched impact strength, IZOD	ISO 180/1A	23°C - 4mm	kJ/m^2	9
THERMAL				
Vicat softening point	ISO 306/B	50N	°C	82
Deflection temperature under load	ASTM D 648	1,8 MPa	°C	81
Coefficient of linear thermal expansion II/⊥	ASTM D 696		10^{-5}/°C	9
COMBUSTIBILITY				
Flammability level test	UL94	1,5 mm	Class	HB
ELECTRICAL				
Surface resistivity	IEC 60093	-	10^{15} ohm	>1.5
Volume resistivity	IEC 60093	-	10^{15} ohm·cm	>7
Dielectric constant	IEC 60250	50 Hz	-	2.5

Tests have been done in 23°C, if it's not marked differently.

3. RECOMMENDATION OF PRINTING

There might be some problems to achieve proper adhesion on glass bed.
Using perforated bed, PVA glue, specialized product (adapted to work with ABS and derivatives), kapton coating or other different way to obtain adhesion (for example 'ABS juice' or hair spray) is recommended.
Similar to other styrene filaments, it is good practice to be slow receiving temperature from freshly applied layers – closed chamber and minimal or turned off the cooling are recommended.
It is necessary to use only with fume extraction or in very well ventilated conditions.

Recommended parameters of printing:

Hotend temperature	230 - 260 °C
Bed temperature	> 80 °C
Print speed	< 100 mm/s

Examples of problems and their solutions:

Problem	Possible cause	Proposed solution
Weak layers adhesion	1) Extruder temperature is too low 2) Too high speed of printing 3) Too efficient cooling	1) Raising the extruder temperature 2) Raising the extruder temperature / reducing the print speed 3) Reducing or turning off the fan responsible for cooling the print
Uneven feeding – losing steps of the feeder / filament sliding on the drive gear	1) Extruder temperature is too low 2) Weak feader pressure	1) Raising the extruder temperature 2) Raising the pressure
Print is peeling of the bed	1) Too low temperature of bed 2) Improperly prepared surface 3) Cooling	1) Raising the bed temperature 2) Degreasing the bed / using another source of adhesion / using perforated bed 3) Giving up the cooling on first stage is recommended

Edge curling	1) Processing shrinkage 2) Too high acceleration	1) Compensation with amount of served filament / selection cooling parameters / changing the chamber temperature 2) Due to high stickiness, high acceleration with sharp edges may cause pulling the edges up

4. SAFETY NOTES

Exhaust fan is recommended.
Air filters in printer is recommended.
HIPS Classic needs to be used only in well ventilated conditions.
Inhaling fumes generated during the printing must be avoided.

Generating fumes during the printing depends mainly on printing temperature. In case of visibly raising emission level, the printing needs to end. Check the hotend temperature and efficiency of the control system before using it next time.

It's forbidden to set fire or exceed recommended printing temperature!

Main ingredient of decomposition is styrene.

Detailed safety information available in SDS.